



Qualification Pack



Stainless Steel Heavy Structure Fabricator

QP Code: ISC/Q0915

Version: 1.0

NSQF Level: 5

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ISC/Q0915: Stainless Steel Heavy Structure Fabricator

Brief Job Description

The individual in this role coordinate and perform stainless steel heavy fabrication process and its associated functions.

Personal Attributes

The person should be patient, organized, team-oriented and have the ability to work for long hours in adverse conditions. They must be able to plan and prioritise tasks effectively and have an eye for detail and quality.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ISC/N0946: Manage and implement health and safety practices at the work place](#)
2. [ISC/N0947: Manage team and others at workplace](#)
3. [ISC/N0948: Understanding Stainless Steel Metallurgy](#)
4. [ISC/N0949: Influence of Elements in Weld Environment](#)
5. [ISC/N0950: Welding Preparation for SS Heavy Fabrication](#)
6. [ISC/N0951: Welding Process of Stainless Steel for Heavy Fabrication as per their Applications - Part I](#)
7. [ISC/N0952: Welding Process for SS for Heavy Fabrication - Part II](#)
8. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Iron and Steel
Sub-Sector	Steel
Occupation	Mechanical Maintenance
Country	India



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NSQF Level	5
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7123.9900
Minimum Educational Qualification & Experience	UG in relevant field (2nd year of 4year UG Degree (BE/ B.Tech) in Metallurgy / Mechanical / Production Engineering) with 6 Months of experience in relevant field OR Completed 2nd year diploma after 12th (in Metallurgy / Material Science / Production Engineering) with 1.5 years of experience in relevant field OR Completed 3 year diploma after 10th (in relevant field Metallurgy / Material Science / Mechanical / Production Engineering) with 2 Years of experience in relevant field OR 12th Class (or equivalent (in relevant field fitter, machinist, welder)) with 3 Years of experience in relevant field
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	27/08/2027
NSQC Approval Date	27/08/2024
Version	1.0
Reference code on NQR	QG-05-IS-02930-2024-V1-ISSSC
NQR Version	1



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ISC/N0946: Manage and implement health and safety practices at the work place

Description

This OS unit is about managing and implementing health, hygiene, safety and sustainable practices for optimising use of resources.

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Maintain health and hygiene
- Manage housekeeping at workplace
- Implement material and energy conservation practices

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** implement safe working practices while dealing with hazards to ensure safety of self and others
- PC3.** ensure that team is following appropriate safety practices while working in and around trenches, elevated places and confined areas
- PC4.** conduct regular checks of the machines with support of the maintenance team to identify potential hazards
- PC5.** organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices
- PC6.** fill daily check sheet to report improvements done and risks identified
- PC7.** ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others
- PC8.** report any identified breaches in health, safety and security policies and procedures to the designated person
- PC9.** ensure that team is following appropriate emergency procedures such as raising alarm, safe evacuation etc.

Maintain health and hygiene

To be competent, the user/individual on the job must be able to:

- PC10.** ensure team is aware about hygiene and sanitation regulations and following them on the shop floor
- PC11.** ensure workplace, equipment, restrooms etc. are cleaned and sanitized regularly
- PC12.** ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace
- PC13.** report advanced hygiene and sanitation issues to appropriate authority



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PC14. report regarding the contagious illness of self or people in close contact

Manage housekeeping at workplace

To be competent, the user/individual on the job must be able to:

PC15. implement appropriate housekeeping methods and practices at workplace

PC16. ensure good housekeeping in order to prevent hazards and accidents

PC17. monitor that that team is storing the material, tools and equipment in the correct location and in good condition after work completion

PC18. ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP

PC19. ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste

Implement material and energy conservation practices

To be competent, the user/individual on the job must be able to:

PC20. identify processes where material and energy/electricity utilization can be optimized

PC21. ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively

PC22. prepare and analyze material and energy audit reports to decipher excessive consumption of material and water

PC23. identify possibilities of using renewable energy and environment friendly fuels

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organisation procedures for health, safety and security, individual role and responsibilities in this context

KU2. the organisation's emergency procedures for different emergency situations and the importance of following the same

KU3. evacuation procedures for workers and visitors

KU4. how and when to report hazards

KU5. potential hazards, risks and threats based on the nature of work

KU6. preventative and remedial actions to be taken in case of exposure to toxic material

KU7. various types of fire extinguisher

KU8. various types of safety signs and their meaning

KU9. appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.

KU10. relevant standards, procedures and policies related to 5S followed in the company

KU11. the various materials used and their storage norms

KU12. efficient utilisation of material and water

KU13. basics of electricity and prevalent energy efficient devices

KU14. common practices of conserving electricity

KU15. categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics

KU16. usage of different colors of dustbins



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KU17. waste management techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain safe and secure working environment</i>	12	21	-	8
PC1. identify hazardous activities and the possible causes of risks or accidents in the workplace	2	3	-	1
PC2. implement safe working practices while dealing with hazards to ensure safety of self and others	2	2	-	1
PC3. ensure that team is following appropriate safety practices while working in and around trenches, elevated places and confined areas	1	2	-	-
PC4. conduct regular checks of the machines with support of the maintenance team to identify potential hazards	2	4	-	1
PC5. organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices	1	2	-	1
PC6. fill daily check sheet to report improvements done and risks identified	1	2	-	1
PC7. ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others	1	2	-	1
PC8. report any identified breaches in health, safety and security policies and procedures to the designated person	1	2	-	1
PC9. ensure that team is following appropriate emergency procedures such as raising alarm, safe evacuation etc.	1	2	-	1
<i>Maintain health and hygiene</i>	6	9	-	4
PC10. ensure team is aware about hygiene and sanitation regulations and following them on the shop floor	2	3	-	1
PC11. ensure workplace, equipment, restrooms etc. are cleaned and sanitized regularly	1	2	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace	1	2	-	1
PC13. report advanced hygiene and sanitation issues to appropriate authority	1	1	-	1
PC14. report regarding the contagious illness of self or people in close contact	1	1	-	-
<i>Manage housekeeping at workplace</i>	6	10	-	4
PC15. implement appropriate housekeeping methods and practices at workplace	2	3	-	1
PC16. ensure good housekeeping in order to prevent hazards and accidents	1	2	-	1
PC17. monitor that that team is storing the material, tools and equipment in the correct location and in good condition after work completion	1	2	-	1
PC18. ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP	1	2	-	1
PC19. ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste	1	1	-	-
<i>Implement material and energy conservation practices</i>	6	10	-	4
PC20. identify processes where material and energy/electricity utilization can be optimized	2	3	-	1
PC21. ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively	1	2	-	1
PC22. prepare and analyze material and energy audit reports to decipher excessive consumption of material and water	2	3	-	1
PC23. identify possibilities of using renewable energy and environment friendly fuels	1	2	-	1
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0946
NOS Name	Manage and implement health and safety practices at the work place
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance, Raw Material Handling , Product Quality Control, Refractory Maintenance, Furnace Operation, Fitter, Cold Rolling Mill Operations, Electrical Maintenance
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



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ISC/N0947: Manage team and others at workplace

Description

This OS unit is about managing team, colleagues/superiors and others, either in own work group or in other work groups within organisation.

Scope

The scope covers the following :

- Communicate effectively with team members
- Interact with supervisor
- Manage appropriate behaviour at work place

Elements and Performance Criteria

Communicate effectively with team members

To be competent, the user/individual on the job must be able to:

- PC1.** manage and co-ordinate with team members to integrate work, as per the workload in order to achieve team goals
- PC2.** implement ways to share information with team members in line with organisational requirements
- PC3.** ensure that work requirements are clearly communicated to the team members through all means including face-to-face, telephonic and written
- PC4.** work in a way that show respect for all team members and customers
- PC5.** carry out commitments made to team members
- PC6.** resolve conflicts within the team members at work to achieve smooth workflow
- PC7.** guide the team members to follow the organisation's policies and procedures
- PC8.** ensure team goals are given preference over individual goals
- PC9.** encourage team members to rectify errors as per feedback and minimize mistakes in future

Interact with supervisor

To be competent, the user/individual on the job must be able to:

- PC10.** report progress on job allocated and team performance to the superiors
- PC11.** escalate problems to supervisors that cannot be handled
- PC12.** report the completed work
- PC13.** train the team members to report completed work and receive feedback on work done

Manage appropriate behaviour at work place

To be competent, the user/individual on the job must be able to:

- PC14.** help PwD team members to overcome the challenges, if asked
- PC15.** ensure team shows sensitivity towards all genders and PwD
- PC16.** adjust communication styles to reflect gender sensitivity and sensitivity towards person with disability



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PC17. ensure that team is following responsible and disciplined behaviours at the workplace

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of effective communication and establishing good working relationships with colleagues and supervisor
- KU2.** different methods of communication as per the circumstances
- KU3.** importance of teamwork in organization and individual success
- KU4.** various components of effective communication
- KU5.** barriers to effective communication
- KU6.** common reasons for interpersonal conflict
- KU7.** what constitutes disciplined behaviour for a working professional
- KU8.** gender concepts, issues & legislation
- KU9.** organisational policies and procedures related to gender equality
- KU10.** challenges faced by PWD and the ways to help them overcome the same

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines/procedures
- GS2.** listen effectively and orally communicate information
- GS3.** ask for clarification and advice from the concerned person
- GS4.** maintain positive and effective relationships with colleagues
- GS5.** evaluate the possible solution(s) to the problem
- GS6.** spot and communicate potential areas of disruptions in the work process and report the same
- GS7.** complete written work with attention to detail
- GS8.** check that the work meets customer requirements



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively with team members</i>	14	24	-	10
PC1. manage and co-ordinate with team members to integrate work, as per the workload in order to achieve team goals	3	5	-	2
PC2. implement ways to share information with team members in line with organisational requirements	3	5	-	2
PC3. ensure that work requirements are clearly communicated to the team members through all means including face-to-face, telephonic and written	2	4	-	2
PC4. work in a way that show respect for all team members and customers	1	2	-	1
PC5. carry out commitments made to team members	1	1	-	1
PC6. resolve conflicts within the team members at work to achieve smooth workflow	1	2	-	-
PC7. guide the team members to follow the organisation's policies and procedures	1	2	-	1
PC8. ensure team goals are given preference over individual goals	1	1	-	1
PC9. encourage team members to rectify errors as per feedback and minimize mistakes in future	1	2	-	-
<i>Interact with supervisor</i>	8	10	-	6
PC10. report progress on job allocated and team performance to the superiors	2	2	-	1
PC11. escalate problems to supervisors that cannot be handled	2	3	-	2
PC12. report the completed work	2	2	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. train the team members to report completed work and receive feedback on work done	2	3	-	2
<i>Manage appropriate behaviour at work place</i>	8	16	-	4
PC14. help PwD team members to overcome the challenges, if asked	2	4	-	1
PC15. ensure team shows sensitivity towards all genders and PwD	2	4	-	1
PC16. adjust communication styles to reflect gender sensitivity and sensitivity towards person with disability	2	4	-	1
PC17. ensure that team is following responsible and disciplined behaviours at the workplace	2	4	-	1
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0947
NOS Name	Manage team and others at workplace
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory, Mechanical Maintenance, Iron Making
Occupation	Mechanical Maintenance, Raw Material Handling , Coke Making, Operations, Iron Making, Cold Rolling Mill Operations, Product Quality Control, Electrical Maintenance, Electronics and Instrumentation Maintenance, Refractory Maintenance, Furnace Operation, Fitter, Steel Making, Pre-fabricated Structure
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N0948: Understanding Stainless Steel Metallurgy

Description

This OS unit is about understanding the basic metallurgy that goes into the making of stainless steel

Scope

The scope covers the following :

- Understanding stainless steel making process
- Learn about different grades of stainless steel and their identification methods
- Learn the welding characteristics of stainless steel

Elements and Performance Criteria

Understanding stainless steel making process

To be competent, the user/individual on the job must be able to:

- PC1.** Carry out practical application of different stainless steel production methods (e.g., electric arc furnace, basic oxygen furnace).
- PC2.** Engage in simulations or visual aids illustrating the sequential steps of stainless-steel making
- PC3.** Understand temperature and timing effects on steel quality.
- PC4.** Highlight quality control measures in stainless steel manufacturing
- PC5.** Handle equipment or tools used in stainless steel refining processes for practical insight

Learn about different grades of stainless steel and their identification methods

To be competent, the user/individual on the job must be able to:

- PC6.** Participating in simulations or visual aids to understand the microstructural make-up of the different grades of SS
- PC7.** Practice in pre-weld and post-weld treatments to understand their influence on stainless steel joint integrity
- PC8.** Reading the results of Spectro and PMI and identify the grade of SS.

Learn the welding characteristics of stainless steel

To be competent, the user/individual on the job must be able to:

- PC9.** various welding techniques suitable for different stainless-steel grades to understand practical applicability
- PC10.** Understand pre-weld and post-weld treatments to understand their influence on stainless steel joint integrity
- PC11.** The function of welding equipment and tools used specifically in stainless steel welding

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the fundamentals of stainless-steel production through the Bessemer and electric arc furnace methods



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- KU2.** Identify the critical stages in stainless steel making: melting, refining, and casting
- KU3.** Recognize the significance of alloying elements and their impact on stainless steel properties.
- KU4.** Grasp the importance of quality control measures in ensuring the desired stainless-steel characteristics.
- KU5.** Explain the refining techniques employed to achieve desired stainless-steel qualities.
- KU6.** Comprehend the role of temperature, chemistry, and timing in stainless steel manufacturing.
- KU7.** Appreciate the environmental considerations and sustainability aspects in stainless steel production.
- KU8.** Understanding the variations in stainless steel grades like austenitic, ferritic, martensitic and duplex SS and their impact on welding properties
- KU9.** Identify suitable welding techniques and parameters for different stainless-steel compositions.
- KU10.** Recognize the trade-offs between strength, corrosion resistance, and formability in different grades of stainless steel.
- KU11.** Recognizing applications and advantages of the different grades of the stainless steels in engineering.
- KU12.** Understand use of methods like Spectro and PMI to identify the grade of SS.
- KU13.** Understand the variations in stainless steel grades and their impact on welding properties.
- KU14.** Recognize the importance of shielding gases and their role in preventing weld contamination.
- KU15.** Identify suitable welding techniques and parameters for different stainless-steel compositions.
- KU16.** Explain the challenges associated with heat-affected zones and distortion in stainless steel welding.
- KU17.** Evaluate the impact of welding defects and their mitigation strategies in stainless steel.
- KU18.** Appreciate the necessity of proper ventilation and safety measures during stainless steel welding processes.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Identify sequential steps in the stainless-steel making process
- GS2.** Understand about the different grades of Stainless Steel
- GS3.** Understand methods to identify the various grades of SS
- GS4.** Select optimal welding techniques and parameters based on SS grades
- GS5.** Apply comprehensive knowledge of shielding gases, defect identification, safety protocols to ensure quality in SS welding



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding stainless steel making process</i>	7	10	-	5
PC1. Carry out practical application of different stainless steel production methods (e.g., electric arc furnace, basic oxygen furnace).	1	2	-	1
PC2. Engage in simulations or visual aids illustrating the sequential steps of stainless-steel making	1	2	-	1
PC3. Understand temperature and timing effects on steel quality.	1	2	-	1
PC4. Highlight quality control measures in stainless steel manufacturing	2	2	-	1
PC5. Handle equipment or tools used in stainless steel refining processes for practical insight	2	2	-	1
<i>Learn about different grades of stainless steel and their identification methods</i>	7	10	-	3
PC6. Participating in simulations or visual aids to understand the microstructural make-up of the different grades of SS	2	3	-	1
PC7. Practice in pre-weld and post-weld treatments to understand their influence on stainless steel joint integrity	2	3	-	1
PC8. Reading the results of Spectro and PMI and identify the grade of SS.	3	4	-	1
<i>Learn the welding characteristics of stainless steel</i>	16	30	-	12
PC9. various welding techniques suitable for different stainless-steel grades to understand practical applicability	7	10	-	4
PC10. Understand pre-weld and post-weld treatments to understand their influence on stainless steel joint integrity	5	10	-	4
PC11. The function of welding equipment and tools used specifically in stainless steel welding	4	10	-	4



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0948
NOS Name	Understanding Stainless Steel Metallurgy
Sector	Iron and Steel
Sub-Sector	
Occupation	Mechanical Maintenance
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024



Qualification Pack

ISC/N0949: Influence of Elements in Weld Environment

Description

This OS unit is about understanding how elements influence the integrity of a weld.

Scope

The scope covers the following :

- Influence of Elements in Weld Environment
- Influence of Alloying on Weldability & Weld Structure

Elements and Performance Criteria

Influence of Elements in Weld Environment

To be competent, the user/individual on the job must be able to:

- PC1.** Engaging in hands-on exercises showcasing hydrogen-induced cracking and sulphur's influence on hot cracking and corrosion in welds.
- PC2.** Analysing practical scenarios highlighting carbon's effects on weld strength and heat-affected zones, and phosphorus's implications for weld quality
- PC3.** Practicing pre-weld and post-weld treatments to understand their influence on stainless steel joint integrity.
- PC4.** Examining case studies illustrating trace elements' alteration of weld characteristics and practicing controlled environments to prevent element-induced weld defects.

Influence of Alloying on Weldability & Weld Structure

To be competent, the user/individual on the job must be able to:

- PC5.** Engage in hands-on exercises to observe the impact of alloy composition on weld heat-affected zones.
- PC6.** Examine samples to identify the influence of alloying elements on weld solidification and defect formation.
- PC7.** Analyse real-life case studies highlighting the trade-offs between alloying for properties and weldability challenges.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the impact of oxygen on weld quality and its role in oxidation and porosity formation.
- KU2.** Understand the effects of nitrogen on weld properties and its contribution to brittleness or strength.
- KU3.** Recognize the significance of hydrogen in welds and its potential for causing cracking.
- KU4.** Identify the role of sulphur in welds and its influence on hot cracking and corrosion resistance.



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- KU5.** Explain the impact of carbon in welds, affecting strength, hardness, and heat-affected zones.
- KU6.** Comprehend the effects of phosphorus on weldability and its contribution to embrittlement.
- KU7.** Evaluate the influence of trace elements like copper, nickel, and chromium in altering weld characteristics.
- KU8.** Understand the effects of specific alloying elements on weld properties, such as strength, ductility, and corrosion resistance
- KU9.** Recognize the role of alloy composition in influencing weld heat-affected zones and microstructural changes.
- KU10.** Explaining how alloying elements affect solidification behaviour and the formation of weld defects.
- KU11.** Explaining the impact of carbon in welds, affecting strength, hardness, and heat-affected zones.
- KU12.** Comprehend the importance of selecting suitable alloys to achieve desired welding outcomes and structural integrity.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understand the effects of various elements in the weld quality.
- GS2.** Practice pre weld and post weld treatments to ensure good welds
- GS3.** Predict and manage weld defects, understand HAZ and microstructural changes influenced by alloying elements
- GS4.** Applying the knowledge in optimizing welding parameters



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Influence of Elements in Weld Environment</i>	15	25	-	10
PC1. Engaging in hands-on exercises showcasing hydrogen-induced cracking and sulphur's influence on hot cracking and corrosion in welds.	4	7	-	3
PC2. Analysing practical scenarios highlighting carbon's effects on weld strength and heat-affected zones, and phosphorus's implications for weld quality	4	7	-	3
PC3. Practicing pre-weld and post-weld treatments to understand their influence on stainless steel joint integrity.	4	7	-	2
PC4. Examining case studies illustrating trace elements' alteration of weld characteristics and practicing controlled environments to prevent element-induced weld defects.	3	4	-	2
<i>Influence of Alloying on Weldability & Weld Structure</i>	15	25	-	10
PC5. Engage in hands-on exercises to observe the impact of alloy composition on weld heat-affected zones.	5	9	-	3
PC6. Examine samples to identify the influence of alloying elements on weld solidification and defect formation.	5	9	-	3
PC7. Analyse real-life case studies highlighting the trade-offs between alloying for properties and weldability challenges.	5	7	-	4
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0949
NOS Name	Influence of Elements in Weld Environment
Sector	Iron and Steel
Sub-Sector	
Occupation	Mechanical Maintenance
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024



Qualification Pack

ISC/N0950: Welding Preparation for SS Heavy Fabrication

Description

This OS is about the methods involved to undertake welding of SS for heavy fabrication

Scope

The scope covers the following :

- Welding Rod selection
- Joint design and Codes and Standards
- Use of Chill bars

Elements and Performance Criteria

Welding Rod selection

To be competent, the user/individual on the job must be able to:

- PC1.** Engage in demonstrations illustrating the impact of different welding rod coatings on weld quality.
- PC2.** Explain the effects of electrode diameter on weld penetration and deposition rates.
- PC3.** Examine various welding rods to identify differences in material composition and usability.
- PC4.** Conduct practical scenarios to select appropriate rods for desired weld characteristics and properties.
- PC5.** Follow safety protocols and precautions associated with electrode selection and usage in welding processes.

Joint Design & Codes & Standards

To be competent, the user/individual on the job must be able to:

- PC6.** Engage in hands-on exercises to create and evaluate different types of weld joints (butt, lap, tee, corner) for various welding scenarios.
- PC7.** Engage in joint preparation, fit-up, and edge geometry's impact on weld quality and strength.
- PC8.** Make groove and fillet weld designs for specific welding applications, interpreting welding symbols, and specifications for communication Conducting practical scenarios to select appropriate rods for desired weld characteristics and properties
- PC9.** Demonstrate safety protocols and precautions associated with welding processes.

Use of Chill Bars

To be competent, the user/individual on the job must be able to:

- PC10.** Engage in hands-on exercises to to place and adjust chill bars to control weld solidification rates and minimize defects.
- PC11.** Practice the proper placement techniques and orientations of chill bars for desired weld characteristics.
- PC12.** Explain effectiveness of chill bars in reducing welding defects and residual stress.

Knowledge and Understanding (KU)



Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Understand the role of welding rods in the welding process and their significance in creating weld joints
- KU2.** Recognizing the impact of electrode coating on welding performance and joint properties
- KU3.** Explaining the influence of electrode diameter and type on weld penetration, deposition rate, and quality.
- KU4.** Comprehending the importance of storage, handling, and preheating procedures for welding rods.
- KU5.** Evaluating the considerations for selecting welding rods to achieve desired weld characteristics and properties.
- KU6.** Appreciating safety measures and precautions related to electrode selection and usage in welding processes.
- KU7.** Understand the significance of proper joint design in welding for structural integrity and load-bearing capacity, including various types of weld joints (butt, lap, tee, corner) and their applications. As per relevant codes & standards
- KU8.** Recognise factors influencing joint design such as preparation, fit-up, edge geometry, material thickness, welding position, and accessibility, ensuring weld quality and strength
- KU9.** Explaining the necessity of groove and fillet weld designs for specific welding applications, considering welding symbols and specifications in communication, and selecting suitable joint designs based on welding process and material properties
- KU10.** Understand the purpose and function of chill bars in the welding process, particularly in controlling weld solidification rates.
- KU11.** Recognize the various types of chill bars and their applications in different welding scenarios.
- KU12.** Evaluating the considerations for selecting appropriate chill bars based on welding process and material requirements.
- KU13.** Apply practical knowledge of chill bar placement, materials, and orientation to optimize weld solidification rates, mitigating defects and residual stress in various welding scenarios. as per AWS D1.6/D1.6M, ASME Section IX, ISO 15614-1, EN 1011-3

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Applying knowledge of welding rod classifications and coatings to make informed selections, optimizing weld quality and joint properties
- GS2.** Utilizing practical understanding in electrode diameter, material compatibility, and handling protocols to ensure precise electrode selection for varied welding conditions and desired outcomes.
- GS3.** Applying practical knowledge of various weld joint types and factors influencing design to create optimized joints for different welding scenarios, ensuring structural integrity and strength
- GS4.** Utilizing hands-on experience in joint preparation, fit-up, edge geometry, and interpreting welding symbols to select and justify suitable joint designs, considering welding processes, material properties, and safety protocols for effective implementation.



Qualification Pack

GS5. Applying practical knowledge of chill bar placement, materials, and orientation to optimize weld solidification rates, mitigating defects and residual stress



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Welding Rod selection</i>	12	20	-	8
PC1. Engage in demonstrations illustrating the impact of different welding rod coatings on weld quality.	2	4	-	2
PC2. Explain the effects of electrode diameter on weld penetration and deposition rates.	2	4	-	2
PC3. Examine various welding rods to identify differences in material composition and usability.	3	4	-	1
PC4. Conduct practical scenarios to select appropriate rods for desired weld characteristics and properties.	3	4	-	1
PC5. Follow safety protocols and precautions associated with electrode selection and usage in welding processes.	2	4	-	2
<i>Joint Design & Codes & Standards</i>	12	20	-	8
PC6. Engage in hands-on exercises to create and evaluate different types of weld joints (butt, lap, tee, corner) for various welding scenarios.	3	5	-	2
PC7. Engage in joint preparation, fit-up, and edge geometry's impact on weld quality and strength.	4	5	-	2
PC8. Make groove and fillet weld designs for specific welding applications, interpreting welding symbols, and specifications for communication Conducting practical scenarios to select appropriate rods for desired weld characteristics and properties	3	7	-	2
PC9. Demonstrate safety protocols and precautions associated with welding processes.	2	3	-	2
<i>Use of Chill Bars</i>	6	10	-	4
PC10. Engage in hands-on exercises to place and adjust chill bars to control weld solidification rates and minimize defects.	2	4	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Practice the proper placement techniques and orientations of chill bars for desired weld characteristics.	2	4	-	1
PC12. Explain effectiveness of chill bars in reducing welding defects and residual stress.	2	2	-	2
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0950
NOS Name	Welding Preparation for SS Heavy Fabrication
Sector	Iron and Steel
Sub-Sector	
Occupation	Mechanical Maintenance
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024



Qualification Pack

ISC/N0951: Welding Process of Stainless Steel for Heavy Fabrication as per their Applications - Part I

Description

This OS is about the welding process of SS for heavy fabrication - for pipes, girders, columns, for weld overlays and welding of steel clad plate.

Scope

The scope covers the following :

- Welding Process of Stainless Steel: Overview of various processes
- Stainless Steel pipe welding
- SS welding for girders
- SS welding for columns
- Weld overlays
- Welding of Steel -Clad plate

Elements and Performance Criteria

Welding Process of Stainless Steel: Overview of various processes

To be competent, the user/individual on the job must be able to:

- PC1.** Engage in hands-on applications of TIG, MIG, Stick, and Laser welding on stainless steel samples
- PC2.** Examine real-life examples to understand process suitability concerning stainless steel grades, thicknesses, and joint configurations.
- PC3.** Practice appropriate welding the selection processes to achieve desired weld quality and productivity
- PC4.** Conduct comparisons of post-weld treatment requirements based on different welding process selections

Stainless Steel Pipe Welding

To be competent, the user/individual on the job must be able to:

- PC5.** Engage in hands-on applications of TIG, MIG, Stick, and Laser welding on stainless steel samples
- PC6.** Examine real-life examples to understand process suitability concerning stainless steel grades, thicknesses, and joint configurations.
- PC7.** Practice appropriate welding the selection processes to achieve desired weld quality and productivity.
- PC8.** Conduct comparisons of post-weld treatment requirements based on different welding process selections

Stainless Steel Welding for Girders

To be competent, the user/individual on the job must be able to:

- PC9.** Engage in hands-on demonstrations to practice specific welding techniques for girder structures, emphasizing joint configurations and welding positions.



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- PC10.** Participate in practical exercises or simulations to comprehend material intricacies, heat input control, and distortion management in girder welding.
- PC11.** Explain the influence of welding parameters like current, speed, and electrode selection on girder weld quality.
- PC12.** Conduct comparisons of post-weld treatment requirements based on different welding process selections

Stainless Steel Welding for Columns

To be competent, the user/individual on the job must be able to:

- PC13.** Engage in hands-on demonstrations to practice specific welding techniques tailored for column structures, emphasizing joint configurations and welding positions.
- PC14.** Carry out Practical exercises or simulations to comprehend material intricacies, heat input control, and distortion management in column welding.
- PC15.** Explain the influence of welding parameters like current, speed, and electrode selection on column weld quality.
- PC16.** Explain how to conduct post-weld treatments and inspections relevant to column welding processes to ensure structural integrity and load-bearing capacity.

Weld Overlays

To be competent, the user/individual on the job must be able to:

- PC17.** Can apply cladding and hard facing techniques for weld overlays on sample stainless steel components.
- PC18.** Explain the impact of deposition rate, dilution, and metallurgical compatibility in different overlay methods
- PC19.** Engage in meticulous surface preparation techniques and joint designs through practical exercises to ensure effective adhesion of overlay materials.
- PC20.** Conduct experiments to adjust welding parameters (heat input, travel speed, deposition rate) and observing their influence on overlay properties.

Welding of Stainless Steel - Clad Plate

To be competent, the user/individual on the job must be able to:

- PC21.** Practice welding techniques specific to stainless steel-clad plate applications.
- PC22.** Carry out practical exercises or simulations to understand challenges and considerations unique to the welding process of stainless-steel clad plates.
- PC23.** Explain the influence of welding parameters like current, speed, and electrode selection on the quality and integrity of the weld in stainless steel-clad plate applications.
- PC24.** Explain how to conduct post-weld treatments and inspections relevant to column welding processes to ensure the reliability of welded stainless steel-clad plates

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the distinct characteristics and applications of various welding processes (TIG, MIG, Stick, Laser) in stainless steel welding as per AWS D1.6/D1.6M
- KU2.** Grasp the differences in process parameters, heat inputs, and shielding methods among these welding techniques



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- KU3.** Recognize the suitability of each welding process for different stainless-steel grades, thicknesses, and joint configurations.
- KU4.** Explain the influence of process selection on weld quality, productivity, and post-weld treatment requirements.
- KU5.** Comprehending the importance of process-specific precautions and best practices in stainless steel welding applications.
- KU6.** Understand the distinct characteristics and applications of various welding processes (TIG, MIG, Stick, Laser) in stainless steel welding as per AWS D1.6/D1.6M
- KU7.** Grasp the differences in process parameters, heat inputs, and shielding methods among these welding techniques
- KU8.** Recognize the suitability of each welding process for different stainless-steel grades, thicknesses, and joint configurations.
- KU9.** Explain the influence of process selection on weld quality, productivity, and post-weld treatment requirements.
- KU10.** Comprehending the importance of process-specific precautions and best practices in stainless steel welding applications
- KU11.** Understanding the critical role of stainless-steel welding in girder fabrication, focusing on joint design and welding positions.
- KU12.** Grasp welding techniques tailored for girder structures, emphasizing efficiency and structural integrity.
- KU13.** Recognize the significance of material selection, heat input, and distortion control in girder welding.
- KU14.** Explain the impact of welding parameters such as current, speed, and electrode choice on girder weld quality.
- KU15.** Comprehending the importance of process-specific precautions and best practices in stainless steel welding applications.
- KU16.** Understand the pivotal role of stainless-steel welding in column construction, emphasizing on Joint design and welding positions.
- KU17.** Grasp welding techniques tailored for column structures, focusing on efficiency, load-bearing capacity, and structural integrity.
- KU18.** Recognize the significance of material selection, heat input control, and distortion management in column welding.
- KU19.** Explain the impact of welding parameters such as current, speed, and electrode choice on column weld quality.
- KU20.** Comprehending the importance of post-weld treatments and inspections specific to column welding.
- KU21.** The role of weld overlays in enhancing corrosion resistance, wear resistance, and overall mechanical properties of stainless-steel components.
- KU22.** Grasp the different techniques like cladding and hard facing used in weld overlays and their specific applications in sectors.
- KU23.** Recognize the nuances of each technique, considering factors like deposition rate, dilution, and metallurgical compatibility.
- KU24.** Recognizes the critical importance of material selection in weld overlays, including considerations for base metal compatibility, desired mechanical properties, and environmental conditions.



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- KU25.** Explains the importance of meticulous surface preparation, ensuring cleanliness and proper profiling, to facilitate effective adhesion of overlay materials.
- KU26.** Understands the significance of joint design parameters, such as groove geometry and preparation, in optimizing the overlay process.
- KU27.** Grasps the role of precise welding parameters, including heat input, travel speed, and deposition rate, in achieving desired overlay properties
- KU28.** Understand the purpose and applications of welding stainless steel-clad plates in diverse industries
- KU29.** Grasp the challenges and considerations unique to the welding process of stainless-steel clad plates.
- KU30.** Recognize the significance of selecting appropriate welding techniques for joining stainless steel cladding with base materials.
- KU31.** Explain the impact of welding parameters on the quality and integrity of the weld in stainless steel- clad plate applications.
- KU32.** Comprehend importance of post-weld inspections and quality assurance in ensuring the reliability of welded stainless steel-clad plates.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Implementing applied understanding of TIG, MIG, Stick, and Laser welding processes to optimize selection parameters, ensuring proficient and high-quality stainless-steel welds.
- GS2.** Utilizing hands-on experience and comparative evaluation to address process-specific factors, including stainless steel grades, joint configurations, and post-weld treatments, aiming for improved welding results
- GS3.** Applying comprehensive knowledge of precise joint preparation and fit-up techniques, along with specialized TIG, MIG, and orbital welding methods for proficient stainless steel pipe welding.
- GS4.** Utilizing hands-on experience and analytical skills to address challenges such as distortion management, material intricacies, and precise parameter control, ensuring superior quality and efficiency in pipe welding processes.
- GS5.** Applying specialized welding techniques, emphasizing joint configurations and precise welding positions for effective stainless steel girder welding. as per AWS D1.6/D1.6M, ASME Section IX, ISO 15614-1, EN 1011-3
- GS6.** Utilizing practical experience and analytical skills to manage material complexities, control heat input, optimize welding parameters, and execute post-weld inspections for superior girder welding outcomes.
- GS7.** Applying advanced welding techniques with a focus on joint configurations and welding positions, ensuring optimized efficiency and load-bearing capacity in stainless steel column welding. as per AWS D1.6/D1.6M, ASME Section IX, ISO 15614-1, EN 1011-3
- GS8.** Utilizing practical experience and analytical skills to manage material intricacies, control heat input, optimize welding parameters, and execute post-weld inspections for superior column welding outcomes.



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- GS9.** Proficiency in applying advanced overlay techniques like cladding and hard facing for enhanced mechanical properties in stainless steel components. Analysis of intricate factors such as deposition rate, dilution, and metallurgical compatibility for informed and effective overlay application as per AWS D1.6/D1.6M, ASME Section IX, ISO 15614-1, EN 1011-3
- GS10.** Informed decision-making in material selection for weld overlays, considering base metal compatibility and desired mechanical properties. Optimization of surface preparation, joint designs, and welding parameters for effective adhesion and superior performance in stainless steel weld overlays.
- GS11.** Demonstrating proficiency in applying welding techniques tailored for stainless steel-clad plate applications, considering the unique challenges and material considerations. as per AWS D1.6/D1.6M, ASME Section IX, ISO 15614-1, ISO 15613 , EN 1011-3, API 582
- GS12.** Utilizing practical experience to optimize welding parameters for stainless steel-clad plates, ensuring the quality and integrity of welds in diverse applications.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Welding Process of Stainless Steel: Overview of various processes</i>	5	10	-	3
PC1. Engage in hands-on applications of TIG, MIG, Stick, and Laser welding on stainless steel samples	2	3	-	1
PC2. Examine real-life examples to understand process suitability concerning stainless steel grades, thicknesses, and joint configurations.	1	2	-	-
PC3. Practice appropriate welding the selection processes to achieve desired weld quality and productivity	1	3	-	1
PC4. Conduct comparisons of post-weld treatment requirements based on different welding process selections	1	2	-	1
<i>Stainless Steel Pipe Welding</i>	5	10	-	3
PC5. Engage in hands-on applications of TIG, MIG, Stick, and Laser welding on stainless steel samples	1	3	-	1
PC6. Examine real-life examples to understand process suitability concerning stainless steel grades, thicknesses, and joint configurations.	1	2	-	-
PC7. Practice appropriate welding the selection processes to achieve desired weld quality and productivity.	2	3	-	1
PC8. Conduct comparisons of post-weld treatment requirements based on different welding process selections	1	2	-	1
<i>Stainless Steel Welding for Girders</i>	5	12	-	4
PC9. Engage in hands-on demonstrations to practice specific welding techniques for girder structures, emphasizing joint configurations and welding positions.	1	3	-	1
PC10. Participate in practical exercises or simulations to comprehend material intricacies, heat input control, and distortion management in girder welding.	1	3	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Explain the influence of welding parameters like current, speed, and electrode selection on girder weld quality.	2	3	-	1
PC12. Conduct comparisons of post-weld treatment requirements based on different welding process selections	1	3	-	1
<i>Stainless Steel Welding for Columns</i>	5	10	-	4
PC13. Engage in hands-on demonstrations to practice specific welding techniques tailored for column structures, emphasizing joint configurations and welding positions.	1	2	-	1
PC14. Carry out Practical exercises or simulations to comprehend material intricacies, heat input control, and distortion management in column welding.	2	2	-	1
PC15. Explain the influence of welding parameters like current, speed, and electrode selection on column weld quality.	1	3	-	1
PC16. Explain how to conduct post-weld treatments and inspections relevant to column welding processes to ensure structural integrity and load-bearing capacity.	1	3	-	1
<i>Weld Overlays</i>	5	4	-	3
PC17. Can apply cladding and hard facing techniques for weld overlays on sample stainless steel components.	1	1	-	-
PC18. Explain the impact of deposition rate, dilution, and metallurgical compatibility in different overlay methods	2	1	-	1
PC19. Engage in meticulous surface preparation techniques and joint designs through practical exercises to ensure effective adhesion of overlay materials.	1	1	-	1
PC20. Conduct experiments to adjust welding parameters (heat input, travel speed, deposition rate) and observing their influence on overlay properties.	1	1	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Welding of Stainless Steel - Clad Plate</i>	5	4	-	3
PC21. Practice welding techniques specific to stainless steel-clad plate applications.	1	1	-	-
PC22. Carry out practical exercises or simulations to understand challenges and considerations unique to the welding process of stainless-steel clad plates.	1	1	-	1
PC23. Explain the influence of welding parameters like current, speed, and electrode selection on the quality and integrity of the weld in stainless steel-clad plate applications.	2	1	-	1
PC24. Explain how to conduct post-weld treatments and inspections relevant to column welding processes to ensure the reliability of welded stainless steel-clad plates	1	1	-	1
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0951
NOS Name	Welding Process of Stainless Steel for Heavy Fabrication as per their Applications - Part I
Sector	Iron and Steel
Sub-Sector	
Occupation	Mechanical Maintenance
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024



Qualification Pack

ISC/N0952: Welding Process for SS for Heavy Fabrication - Part II

Description

This OS is about processes involved in welding SS for heavy fabrication including welding dissimilar metals, post weld operations, DT & NDT methods and preparation of sample WPS & PQR

Scope

The scope covers the following :

- Welding dissimilar metals
- Post weld cleaning and finishing
- Quality assessment through DT & NDT
- Preparation of sample WPS and PQR

Elements and Performance Criteria

Welding Dissimilar Metals

To be competent, the user/individual on the job must be able to:

- PC1.** Engage in hands-on demonstrations to practice welding techniques specific to dissimilar metal combinations.
- PC2.** Participate in practical exercises or simulations to understand the challenges and considerations unique to welding
- PC3.** Conduct experiments to observe the impact of thermal expansion and contraction on dissimilar metal welds
- PC4.** Apply suitable welding techniques for joining dissimilar metal types, emphasizing material Compatibility.
- PC5.** Analyse the formation of intermetallic compounds during welding and their influence on the mechanical properties of dissimilar metal welds.

Post Weld Cleaning & Finishing

To be competent, the user/individual on the job must be able to:

- PC6.** Engage in hands-on demonstrations to practice various post-weld cleaning methods, including mechanical, chemical, and ultrasonic cleaning techniques.
- PC7.** Participate in practical exercises or simulations to understand the impact of proper surface finishing through grinding, polishing, and shot blasting on welded components.
- PC8.** Conduct experiments to observe the removal of contaminants like oxides and scales and evaluating the effectiveness of different cleaning methods.
- PC9.** Apply knowledge of material properties and welding conditions to select appropriate post-weld cleaning techniques for specific applications.
- PC10.** Gaining practical insights into adhering to industry standards and guidelines for post-weld cleaning and finishing in real-world scenarios.

Quality Assessment through DT & NDT

To be competent, the user/individual on the job must be able to:



Qualification Pack

- PC11.** Engaging in practical exercises to perform DT techniques such as tensile testing (ASTM E8), impact testing (ASTM E23), and hardness testing (ASTM E10) following international standards
- PC12.** Participating in live demonstrations of NDT methods like radiography (ASTM E94), ultrasonic testing (ASTM E213), and magnetic particle testing (ASTM E1444), aligning with ASTM, ASME Section V, and ISO standards.
- PC13.** Applying knowledge to choose appropriate DT or NDT methods based on material characteristics, weld configuration, and industry standards (AWS D1.1/D1.1M, ASME Section V, ISO).
- PC14.** Practicing quality assurance and control measures during DT and NDT processes, in accordance with ISO 9712, ASNT SNT-TC-1A, and national standards such as DIN, PN, and IS.
- PC15.** Developing skills in interpreting and analysing results obtained from DT and NDT methods, ensuring compliance with relevant codes and standards for superior quality assessment

Preparation of sample WPS and PQR

To be competent, the user/individual on the job must be able to:

- PC16.** Reading a WPS and pick up the essential and non-essential variables mentioned.
- PC17.** Actively help in preparation of sample WPS

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the challenges and considerations associated with welding dissimilar metals.
- KU2.** Grasp the significance of material compatibility and selection in dissimilar metal welding.
- KU3.** Recognize the impact of thermal expansion and contraction on dissimilar metal welds
- KU4.** Explain suitable welding techniques for joining different metal types.
- KU5.** Comprehend the role of intermetallic compounds and their influence on the mechanical properties of dissimilar metal welds.
- KU6.** Understanding the critical role of post-weld cleaning in eliminating contaminants like oxides, scales, and weld spatter to enhance the structural integrity and corrosion resistance of welded joints.
- KU7.** Grasping various cleaning methods such as mechanical cleaning, chemical cleaning, and ultrasonic cleaning, tailored to specific welding processes and materials.
- KU8.** Recognizing the importance of choosing appropriate cleaning techniques based on the material type and welding method to avoid surface damage.
- KU9.** Understanding how proper surface finishing techniques contribute to improved aesthetics, corrosion resistance, and fatigue strength of welded components.
- KU10.** Explaining the role of finishing processes like grinding, polishing, and shot blasting in achieving desired surface characteristics.
- KU11.** Recognizing the influence of residual stresses and heat-affected zone (HAZ) conditions on post-weld cleaning considerations.
- KU12.** Understanding how specific cleaning techniques may be selected based on the material properties and welding conditions to mitigate potential issues.
- KU13.** Comprehending the importance of adhering to industry standards and guidelines for post-weld cleaning and finishing to ensure compliance and quality in diverse.



Qualification Pack

- KU14.** Understanding the fundamental principles behind both DT and NDT methods, following standards such as ASTM E165, ISO 17636, ASME Section V, and AWS
- KU15.** Grasping the applications and limitations of common DT techniques such as tensile testing (ASTM E8), impact testing (ASTM E23), and hardness testing (ASTM E10), following ASTM and ISO standards etc
- KU16.** Understanding the significance of NDT methods, including radiography (ASTM E94), ultrasonic testing (ASTM E213), and magnetic particle testing (ASTM E1444), following ASTM, ISO, and ASME Section V standards.
- KU17.** Explaining how the choice between DT and NDT methods is influenced by factors such as material type, weld configuration, and industry standards, including AWS D1.1/D1.1M, ASME Section V, and ISO standards.
- KU18.** Comprehending the critical role of quality assurance and control through DT and NDT processes to ensure compliance with industry standards and project specifications, including ISO 9712, ASNT SNT-TC-1A, and relevant national standards such as DIN (Deutsche Institute fur Normung), PN (Polski Normal), and IS (Indian
- KU19.** Explaining how NDT techniques contribute to identifying flaws and irregularities within the welded joint
- KU20.** Understand the meaning of pWPS, WPS, PQR and the steps involved therein.
- KU21.** Understand the variables - essential, non-essential, supplementary variables in preparing a WPS
- KU22.** Planning and preparation of WPQT.
- KU23.** Process of welder qualification.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrating advanced welding techniques for dissimilar metal combinations. Analysing challenges for optimal joint integrity.
- GS2.** Utilizing practical experience to optimize material compatibility. Applying suitable welding techniques for desired mechanical properties. (as per AWS D17.1/D17.1M, AWS D1.1/D1.1M, ASME Section IX, ISO 9606-1, ISO 15614-1 etc)
- GS3.** Applying advanced post-weld cleaning methods, emphasizing tailored approaches for diverse materials and welding processes, aligning with industry standards including ISO 8501-1 and AWS C3.7. and others
- GS4.** Utilizing hands-on experience and analytical skills to optimize surface finishing through grinding, polishing, and shot blasting, ensuring superior aesthetics, corrosion resistance, and fatigue strength in welded components.
- GS5.**
 - Achieving proficiency in applying a range of Destructive Testing (DT) and Non-Destructive Testing
 - (NDT) techniques, such as ultrasonic testing (ASTM E213), radiography (ASTM E94), and tensile testing (ASTM E8), aligning with international standards. as per AWS D1.6/D1.6M, ASME Section IX, ISO 15614-1, EN 1011-3
- GS6.** Demonstrating strategic implementation of quality assurance practices in DT and NDT processes, ensuring compliance with industry standards (AWS D1.1/D1.1M, ASME Section V, ISO) and fostering superior weld quality assessments.



Qualification Pack

GS7. Understanding and Preparation of sample WPS and PQR as per requirement



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Welding Dissimilar Metals</i>	6	15	-	5
PC1. Engage in hands-on demonstrations to practice welding techniques specific to dissimilar metal combinations.	2	3	-	1
PC2. Participate in practical exercises or simulations to understand the challenges and considerations unique to welding	1	3	-	1
PC3. Conduct experiments to observe the impact of thermal expansion and contraction on dissimilar metal welds	1	3	-	1
PC4. Apply suitable welding techniques for joining dissimilar metal types, emphasizing material Compatibility.	1	3	-	1
PC5. Analyse the formation of intermetallic compounds during welding and their influence on the mechanical properties of dissimilar metal welds.	1	3	-	1
<i>Post Weld Cleaning & Finishing</i>	8	15	-	5
PC6. Engage in hands-on demonstrations to practice various post-weld cleaning methods, including mechanical, chemical, and ultrasonic cleaning techniques.	2	3	-	1
PC7. Participate in practical exercises or simulations to understand the impact of proper surface finishing through grinding, polishing, and shot blasting on welded components.	2	3	-	1
PC8. Conduct experiments to observe the removal of contaminants like oxides and scales and evaluating the effectiveness of different cleaning methods.	1	3	-	1
PC9. Apply knowledge of material properties and welding conditions to select appropriate post-weld cleaning techniques for specific applications.	2	3	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Gaining practical insights into adhering to industry standards and guidelines for post-weld cleaning and finishing in real-world scenarios.	1	3	-	1
<i>Quality Assessment through DT & NDT</i>	8	10	-	5
PC11. Engaging in practical exercises to perform DT techniques such as tensile testing (ASTM E8), impact testing (ASTM E23), and hardness testing (ASTM E10) following international standards	1	2	-	1
PC12. Participating in live demonstrations of NDT methods like radiography (ASTM E94), ultrasonic testing (ASTM E213), and magnetic particle testing (ASTM E1444), aligning with ASTM, ASME Section V, and ISO standards.	1	2	-	1
PC13. Applying knowledge to choose appropriate DT or NDT methods based on material characteristics, weld configuration, and industry standards (AWS D1.1/D1.1M, ASME Section V, ISO).	2	2	-	1
PC14. Practicing quality assurance and control measures during DT and NDT processes, in accordance with ISO 9712, ASNT SNT-TC-1A, and national standards such as DIN, PN, and IS.	2	2	-	1
PC15. Developing skills in interpreting and analysing results obtained from DT and NDT methods, ensuring compliance with relevant codes and standards for superior quality assessment	2	2	-	1
<i>Preparation of sample WPS and PQR</i>	8	10	-	5
PC16. Reading a WPS and pick up the essential and non-essential variables mentioned.	5	5	-	3
PC17. Actively help in preparation of sample WPS	3	5	-	2
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0952
NOS Name	Welding Process for SS for Heavy Fabrication - Part II
Sector	Iron and Steel
Sub-Sector	
Occupation	Mechanical Maintenance
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024



Qualification Pack

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment



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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms



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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services



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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70



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(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ISC/N0946.Manage and implement health and safety practices at the work place	30	50	-	20	100	5
ISC/N0947.Manage team and others at workplace	30	50	-	20	100	5
ISC/N0948.Understanding Stainless Steel Metallurgy	30	50	-	20	100	15
ISC/N0949.Influence of Elements in Weld Environment	30	50	-	20	100	15
ISC/N0950.Welding Preparation for SS Heavy Fabrication	30	50	-	20	100	15
ISC/N0951.Welding Process of Stainless Steel for Heavy Fabrication as per their Applications - Part I	30	50	-	20	100	20
ISC/N0952.Welding Process for SS for Heavy Fabrication - Part II	30	50	-	20	100	20
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	5
Total	230	380	-	140	750	100



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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.